REALISING A MORPHISM BETWEEN THE CLOSED FIBRES BY A CORRESPONDENCE IN THE BIG ETALE SITE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file etale-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument: the openness of the graph and the choice of W are both inside the counted span.

Editorial note. The label lemma-closed-immersion-almost-full occurs in no other staged Stacks chapter, so no cross-chapter restatement had to be weighed for this card.

Editorial note. The single labelled reference occurring inside the counted spans points to a chapter of the Project not staged in this card's source tree; it is carried in assembly.external_references as a non-load-bearing navigation pointer recorded at the invoking line.

Editorial note. The preamble is the chapter's own native preamble excerpt already used for this source: the Xy-pic setup (native preamble lines 14-18) and the native theorem declarations with the mathematical macros (native preamble lines 154-217).

Allowed standard prerequisites (editorial).

- Scheme U_Z = U x_X Z attached to a scheme U etale over X and a closed immersion i : Z -> X. Same chapter, the notation U_Z for the base change of an object of X_etale along the closed immersion. Supplies the source and target of the given morphism h and the closed subscheme M_Z of the product.
- Graph of a morphism and its openness for an etale projection. Same chapter together with the chapter on Etale Morphisms, the fact that the graph of a morphism to an etale scheme is the image of a section, hence open. Supplies the openness of the graph Gamma_h inside M_Z, which is the crux of the proof.
- Open subscheme with prescribed closed fibre. Same chapter, the standard fact that an open subscheme of the closed fibre M_Z can be lifted to an open subscheme W of M. Supplies the open subscheme W of M = U x_X U' whose closed fibre is Gamma_h.

Source reference. Original proposition 024T: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 6614–6614. Cross-chapter navigation pointer invoked at proof lines 6612-6614, where the author uses the properties of sections of an etale morphism, in particular that the image of a section is open. The target chapter is not staged in this card's source tree, so the pointer records the invoking line; it only resolves the reference for the reader, and no bibliography citation is involved.

Date: stacks-project:etale-cohomology.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Lemma 0.1. *Let $i : Z \to X$ be a closed immersion of schemes. Let U, U' be schemes étale over X. Let $h : U_Z \to U'_Z$ be a morphism over Z. Then there exists a diagram*

$$U \xleftarrow{a} W \xrightarrow{b} U'$$

in $X_{étale}$ such that $a_Z : W_Z \to U_Z$ is an isomorphism and $h = b_Z \circ (a_Z)^{-1}$.

Proof. Consider the scheme $M = U \times_X U'$. The graph $\Gamma_h \subset M_Z$ of h is open. This is true for example as Γ_h is the image of a section of the étale morphism $\mathrm{pr}_{1,Z} : M_Z \to U_Z$, see Étale Morphisms, Proposition 024T. Hence there exists an open subscheme $W \subset M$ whose intersection with the closed subset M_Z is Γ_h. Set $a = \mathrm{pr}_1|_W$ and $b = \mathrm{pr}_2|_W$. □